

Exam Date & Time: 14-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

ESSENTIALS OF QUANTUM COMPUTING [CSE-40B1]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the concept of complex number multiplication and identify the underlying components of rotation and scaling in the complex plane. Determine the relationship between the algebraic representation of complex numbers and their geometric transformations with a neat sketch. (5)
 - A)
 - B)
- 2) Evaluate the responsibilities of the control processor plane in a quantum computer. How does it manage quantum operations? (5)
- 3) Analyze the concept of quantum superposition using the example of a qubit. Explain how this differs from classical binary systems? (5)

A)

B)

- A) A single qubit is initialized in state $|0\rangle$ and $|1\rangle$. Analyze the effect of applying a Hadamard gate followed by a Pauli-Z gate, and then another Hadamard gate on $|0\rangle$ and $|1\rangle$ separately. What is the final state, and what does this reveal about phase shift operations? (5)

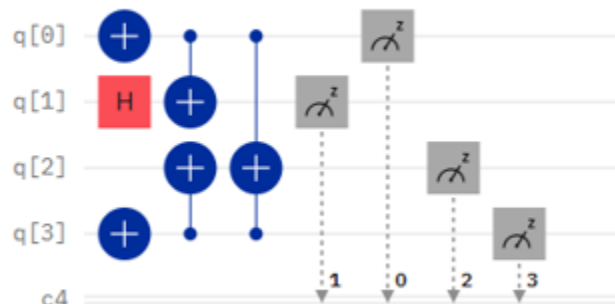
- 3) Determine whether U is a unitary matrix,

A)

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & i \\ 0 & \sqrt{2} & 0 \\ i & 0 & 1 \end{bmatrix} \quad (5)$$

B)

Analyze the given quantum circuit and find the quantum state at each level.



(5)

- 4) Analyze how the choice of basis affects the interpretation of a quantum operator. Consider a linear operator B acting on a 2D Hilbert space with the following properties: $B|0\rangle=|0\rangle$, $B|1\rangle=-|1\rangle$
- A) Find the matrix representation of B in the computational basis $\{|0\rangle,|1\rangle\}$. Express the action of B in the Hadamard basis $\{|+\rangle,|-\rangle\}$ and compare the operator's behavior in both bases. (5)
- B) Explain controlled gates and evaluate the effectiveness of using different controlled gates (CNOT, CZ, Toffoli) for generating entanglement and implementing conditional logic in quantum circuits. Compare their behavior in terms of gate complexity, entanglement capability, and circuit efficiency. (5)
- 5) Analyze the action of a phase gate (S) and T gate on the Bloch sphere. What types of rotations do they induce and how do they affect the relative phase between basis states? Apply these gates on $|+\rangle$ and show the final state. (5)
- A)
- B) Compare the major types of physical qubits used in quantum computing, and how do their characteristics affect scalability and stability? (5)

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